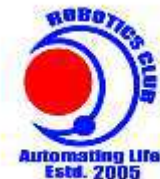


Introduction to Git and GitHub



Speaker:
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Susan Ghimire

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1. GitHub Actions

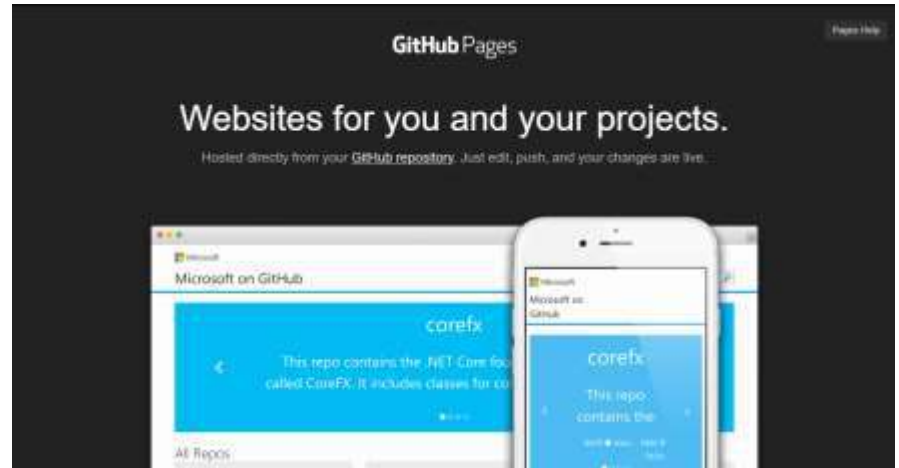


1. VS Code Extensions



What is GitHub Pages?

- A **static site hosting service** that takes HTML, CSS, and JavaScript files straight from a repository on GitHub, optionally runs the files through a build process, and publishes a website.
- You can use GitHub Pages to host a website about yourself, your organization, or your project directly from a repository on GitHub.com.



More about GitHub Pages

- It's **FREE**
- One site per GitHub account and organization (<https://username.github.io>)
- Unlimited project sites.
(<http://username.github.io/repository>)



Usage Limits



- GitHub Pages source repositories have a recommended limit of **1GB**.
- GitHub Pages sites have a ***soft*** bandwidth limit of **100GB per month**.
- GitHub Pages sites have a ***soft*** limit of **10 builds per hour**.

What is a GitHub Actions?

- A continuous integration and continuous delivery (CI/CD) platform to automate your build, test, and deployment pipeline.
- You can create workflows that build and test every pull request to your repository, or deploy merged pull requests to production.



GitHub Actions

Working of GitHub Actions

- You can configure a GitHub Actions *workflow* to be triggered when an *event* occurs in your repository.
- Your workflow contains one or more *jobs* which can run in sequential order or in parallel.
- Each job will run inside its own virtual machine *runner*, or inside a container, and has one or more *steps*



Components of GitHub Actions

1. Workflows

A workflow is a configurable automated process that will run one or more jobs. You must create a YAML file to define your workflow configuration.

Syntax for YAML: [Here](#)

1. Events

An event is a specific activity in a repository that triggers a workflow run.



Components of GitHub Actions

3. Jobs

A job is a set of *steps* in a workflow that execute on the same runner. Each step is either a shell script that will be executed, or an *action* that will be run.

3. Runners

A runner is a server that runs your workflows when they're triggered. Each runner can run a single job at a time. GitHub provides Ubuntu Linux, Microsoft Windows, and macOS runners to run your workflows.



Usage Limits for Free Plan

- Each job in a workflow can run for up to 6 hours of execution time.
- Each workflow run is limited to 72 hours.
- You can execute up to 1000 API requests in an hour across all actions within a repository.
- You can run upto 20 concurrent jobs in your account

30 Days
free trial

ADD Credit/Debit
card to
proceed

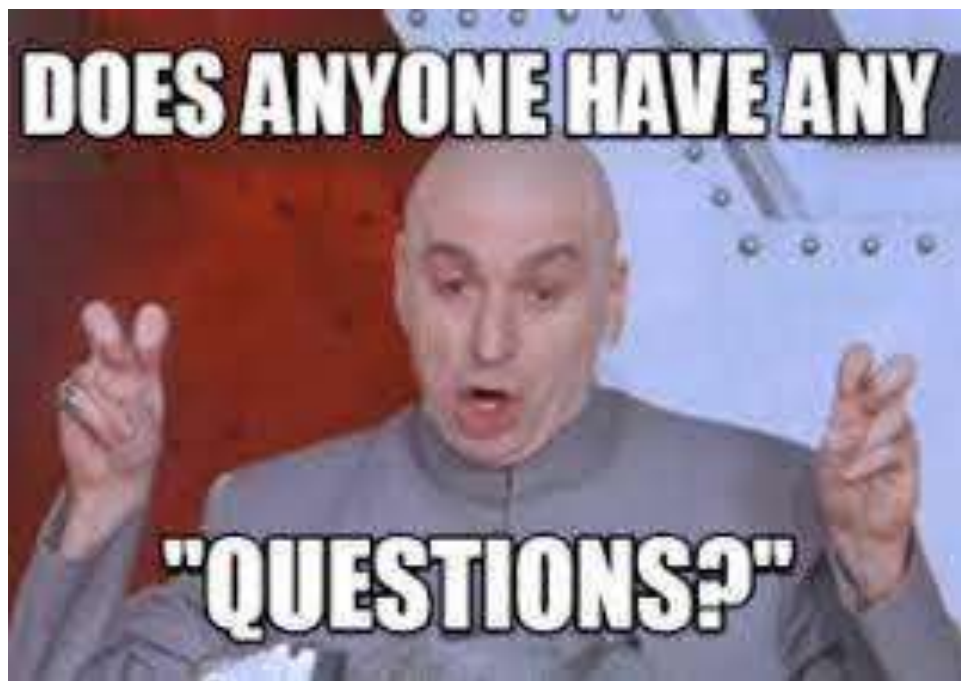


Cron Expression



```
# * * * * * command to execute
```

```
# |  
# |  
# |  
# |  
# |____ day of week (0 - 7)  
# |____ month (1 - 12)  
# |____ day of month (1 - 31)  
# |____ hour (0 - 23)  
# |____ min (0 - 59)
```



Assignment

If you're facing any difficulties:

Contact us at:

Facebook: [KEC I.T. Club](#)

[KEC Robotics Club](#)

